

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · PHYSICS

PHYSICS 6032/2
PAPER 2 November 2023
2 hours 15 minutes

Candidate Name: _____

Centre Number: _____ **Candidate Number:** _____

TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES

Data and Formula booklet provided. Electronic calculator permitted.

Answer all questions. Show clear mathematical working with derived physical units.

SECTION A

1. A mass of 0.45 kg oscillates with SHM at the end of a vertical spring. The displacement $x = 0.080 \sin(15t)$ meters. (a) State the amplitude and angular frequency of the motion. (b) Calculate the maximum velocity and maximum acceleration of the mass. (c) Determine the total mechanical energy of the oscillator. *[Simple Harmonic Motion & Circular Motion]* [8]
- (a) State the amplitude and angular frequency. [2]
- (b) Calculate maximum speed and acceleration. [3]
- (c) Determine total oscillatory energy. [3]